

TerraHawk FPV v1.0 Assembly Manual

Thank you for purchasing the TerraHawk FPV Digital Kit!

For the best results, follow the assembly manual, take your time, stay safe and most importantly - enjoy the build process.

Need Support?

Need assistance or eager to share your progress? Join our Discord community:

<https://discord.gg/nhsGWzRsYf>

Resources:

[3D Printing a TerraHawk: Beginners Guide](#)

[ArduPilot QuadPlane Setup: Beginners Guide](#)

[Recommended Electronics List](#)

Tools and consumables for a TerraHawk FPV build:

- FDM 3D printer, desktop class
- Bambu Lab X1 / P1 / A1 or similar
- Prusa i3 MK3S+ / MK4
- Creality Ender 3 (Pro / V2 / V3)
- Anycubic Kobra 2 (base / Pro)
- ~1 kg of lightweight filament (PLA Aero) and ~250 g of durable filament (PETG)
- CA glue (medium), hot glue, double-sided tape, thread-locker
- M3 screws and nuts, including long ones of 30 mm or more. M2 screws for the camera
- Fine-tooth saw for carbon tube, 3 mm drill bit, hand screwdriver
- Servo tester, servo extension leads, control horns and rods, motor wire extensions
- Flashlight, tape, basic hand tools
- Protective gear: safety goggles, respiratory mask, gloves

#1 ORDER YOUR PARTS AND FILAMENT

Do this before you print anything.

A full airframe can take days of printer time. Electronics and carbon spars take roughly as long to arrive. Order on day one and the two happen in parallel.

Filament

What	How much	Used for
Lightweight foaming filament (PLA Aero or LW-PLA)	~1 kg	Fuselage, wings, elevons, nose cone
PETG	~250 g	Spar plates, mounts, hardware interfaces

- **Starter builds:** PLA Aero + PETG. Basic, lighter. Alternatively LW-PLA.
- **Performance builds:** ABS-GF + PETG-CF. Engineering-grade, heavier, enclosure required.
- **Heat resistant builds:** ASA Aero + PETG. Difficult to print, light, heat resistant.

Brands and settings are in **#2 Print the Airframe**.

Carbon spars

Order **2m lengths**, if possible. It cuts waste to almost nothing and costs just a bit more to deliver.

Order	Cuts into	Left over
2m × 8mm pultruded	Rear fuselage spar + both wing spars	50mm
2m × 14mm woven	Front fuselage spar + both VTOL spars	760mm
1m × 6mm pultruded	Both elevon spars	300mm

The 8mm rod has 50mm of margin across three cuts.

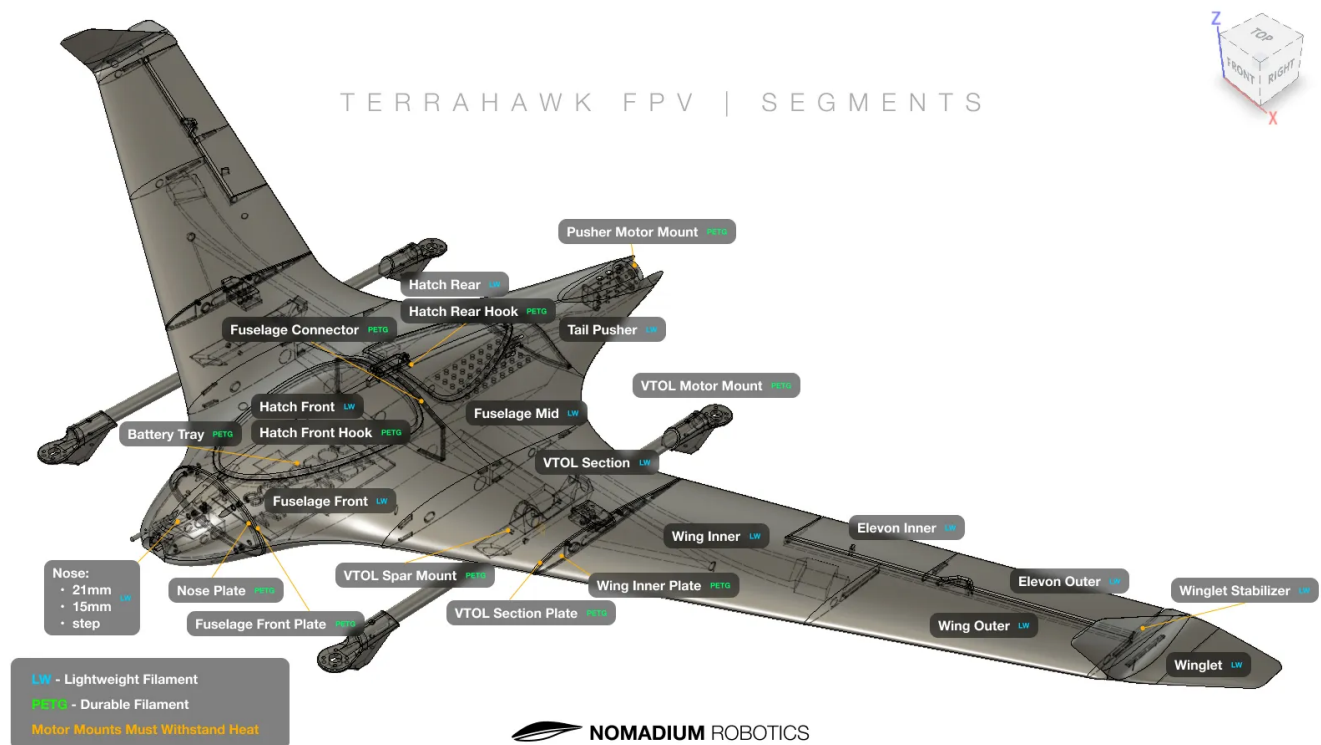
Measure twice, mark all three cuts before you cut any of them, and allow for the width of your blade. The full cut list is in **#3 Assemble the Airframe**.

Electronics

Every flight-critical component, tested by us:

[Recommended Electronics List](#)

#2 PRINT THE AIRFRAME



Segments Overview. TerraHawk is cut into segments so it fits most desktop 3D printers. Segments print in two filament types, a lightweight one and a durable one, as shown in the image (LW and PETG).

Segments are glued together with a glue that suits your filament. For most builds, CA glue (superglue) is enough. It melts and fuses non-engineering filaments together.

TerraHawk has quick-release wings and a swappable nose cone. They separate at durable plates, which must never be glued to each other.

Building the pusher-only variant? The STL kit includes a non-VTOL section for it. This manual covers the VTOL build. The pusher-only variant is for experienced builders.

Never printed an aircraft before? Start here instead. It explains every setting below, and how to read a test print.

[3D Printing a TerraHawk: Beginners Guide](#)

Print Settings

Active-foaming filaments recommended. **Flow is a ratio, not a percentage.**

Filament	Nozzle	Flow ratio	Bed	Part cooling
Bambu Lab PLA Aero (recommended)	220 °C (default)	0.60 (default)	55 °C, textured PEI	~30%
ColorFabb LW-PLA	230 °C	0.65	60 °C, textured PEI	~20%
eSUN LW-PLA	215 °C	0.57	55 °C, textured PEI	~35%

Filament	Nozzle	Flow ratio	Bed	Part cooling
ColorFabb LW-PLA-HT	220 °C (210 first layer)	0.60	60 °C, textured PEI	~20%
ColorFabb LW-ASA	240 °C	0.58	100 °C, high-temp plate + glue	10-50%
Bambu Lab ASA Aero	245 °C	0.61	90 °C, high-temp plate + glue	10-50%

- **Starter builds:** PLA Aero + PETG. Basic, lighter. Alternatively LW-PLA.
- **Performance builds:** ABS-GF + PETG-CF. Engineering-grade, heavier, enclosure required.
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The two ASA rows run hotter than the PLAs because ASA will not bond layers properly below about 240 °C. They also need an enclosed printer.

These run cooler than the foaming profiles you will find online, and that is deliberate.

Public profiles are tuned for **ultra-light gliders**, where every gram matters more than strength. They push 250 to 267 °C to get maximum foaming.

TerraHawk is a ~2kg VTOL. It carries five motors, a battery and an FPV payload, and it lands on printed feet.

Cooler means less foaming, which means denser and stronger. The temperatures above are the strength-to-weight optimum for this aircraft, not the lightest possible print.

PLA Aero at 220 °C and 0.60 flow is our validated setting. Start there if you can source it. The other five rows are derived from it, scaled to each filament's own usable window. Good starting points, not measured results.

Slicer Profile

Start from a stock 0.4mm nozzle profile with default filament settings, then change these.

Setting	Lightweight parts	PETG parts
Layer height (default)	0.2mm	0.2mm
Walls	1	3
Infill	Cubic, 4%	Gyroid, 10%
Avoid Crossing Walls	ON	-
Print speed	~35 mm/s (or default)	Stock profile
Brims (wings, elevons)	Outer, 5mm, 0.07 distance	-

Slow down by height (Bambu Studio). Optional, but recommended: 100% at the 1st layer down to 25% at the last. Wings and elevons are tall and thin, and vibration costs you print quality toward the tip. Especially useful for bed-slinger printers like the Bambulab A1.

Five Things That Will Bite You

Avoid Crossing Walls must be ON. Otherwise the nozzle oozes into the spar holes on every travel move and the spars barely fit.

Flow ratio is not a typo. 0.6 means 60%. Foaming filament expands in the nozzle, so you push less of it. Some LW-PLA variants come pre-foamed - check recommendations.

Dry the filament. 55 °C for 8 hours. Wet foaming filament foams unevenly, and that is a wing with soft spots in it.

Turn off flow dynamics / pressure advance calibration for foaming filament. It cannot measure a material that expands after it leaves the nozzle.

One wall means one wall. Do not add a perimeter "for strength". The airframe is designed around a single-wall shell and 4% infill. Not applicable to non-VTOL builds.

Part Orientation

Print one part at a time for the best results.

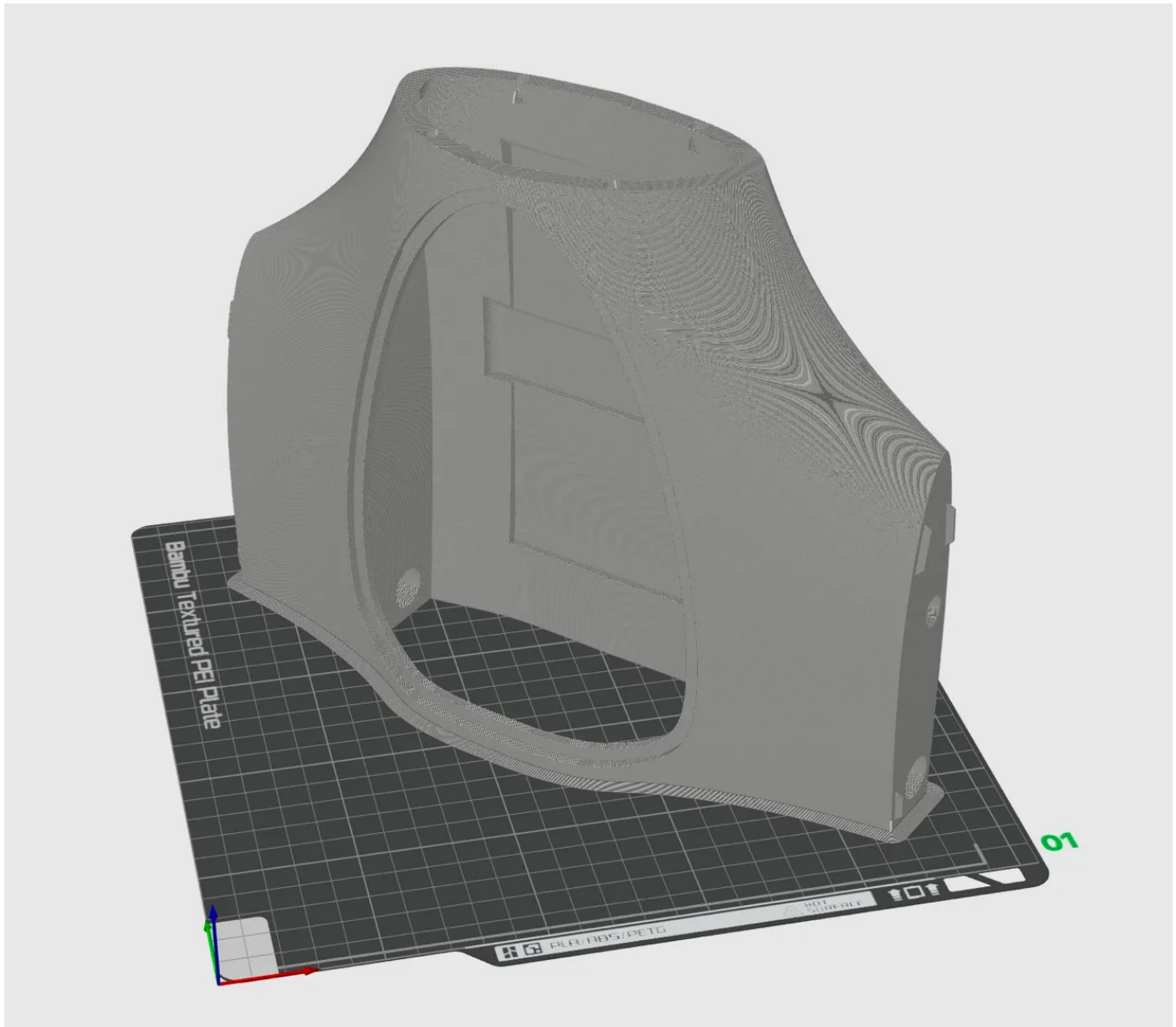
General rules. Every segment is cut with a flat mating face, and that face is designed to sit on the bed. The cuts already account for layer strength, so do not rotate parts to "improve" it.

Oriented correctly, few parts need **supports**. Disable them for parts that do not require them - listed below:

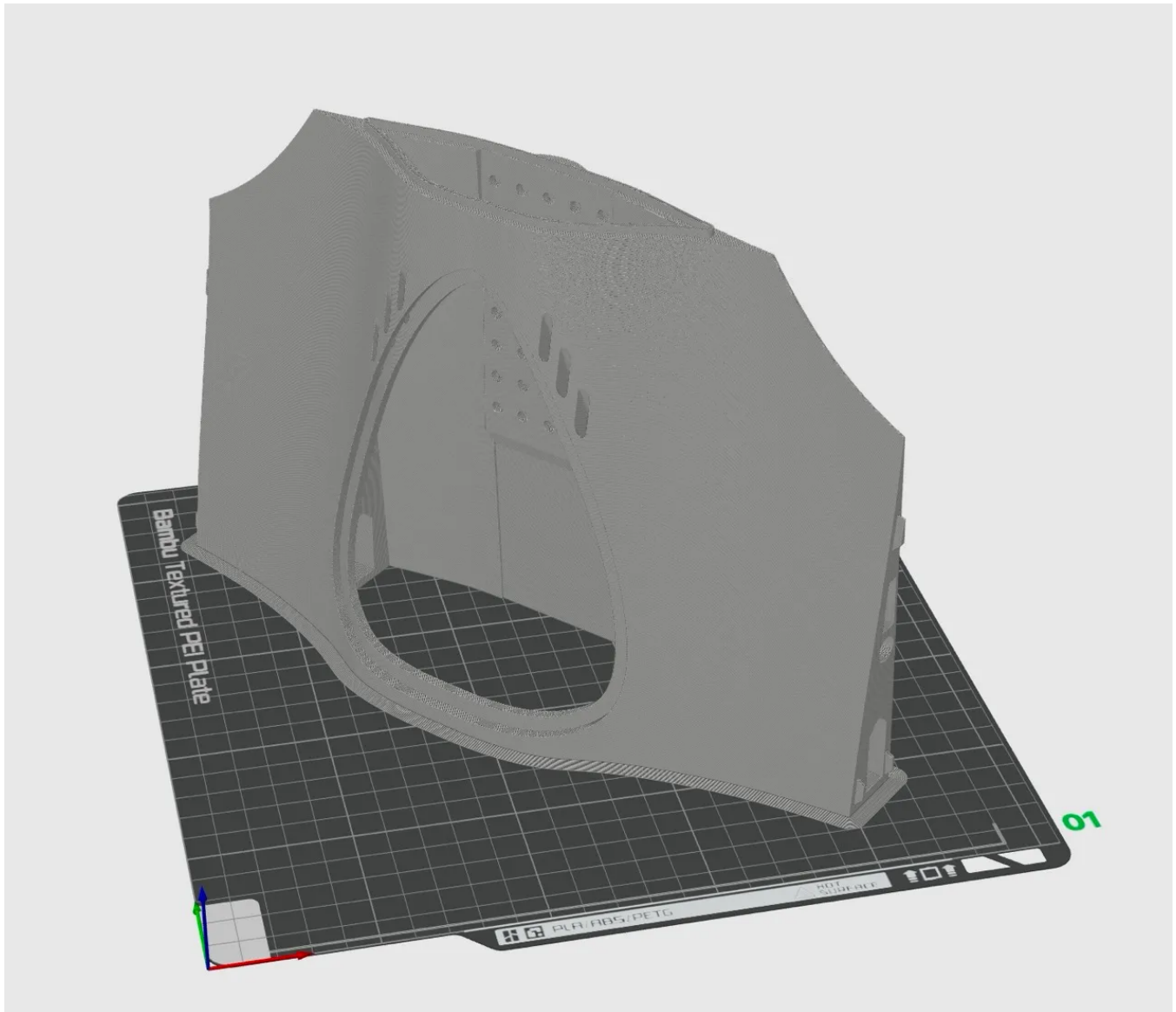
Never printed a part with Lightweight filament (LW)? Try a smaller part first like the Winglet.

Sharp edges serve aerodynamic purposes, but are also prone to cutting human skin. Handle them with care. All parts should be kept away from children and animals.

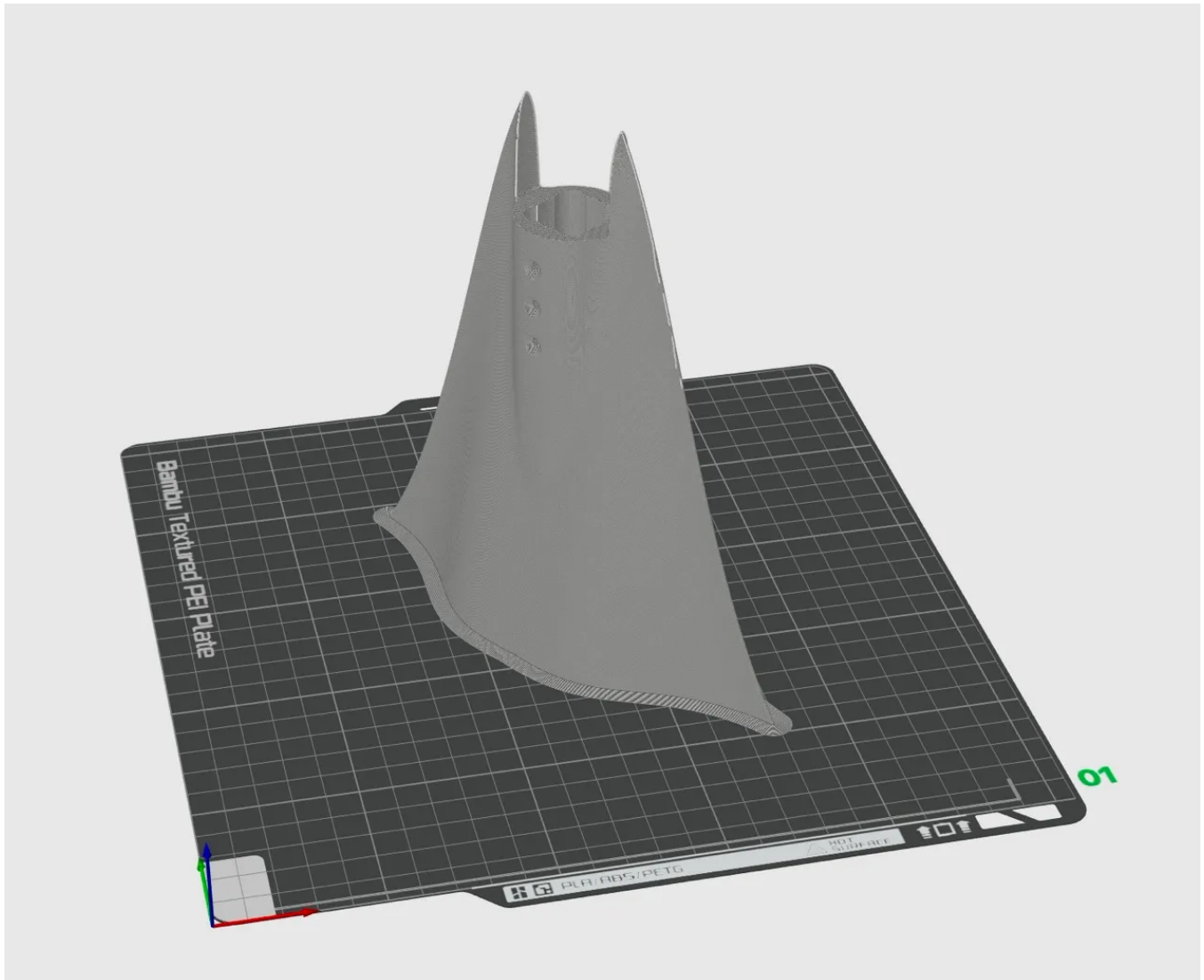
Fuselage Front | Lightweight filament (LW)



Fuselage Mid | LW

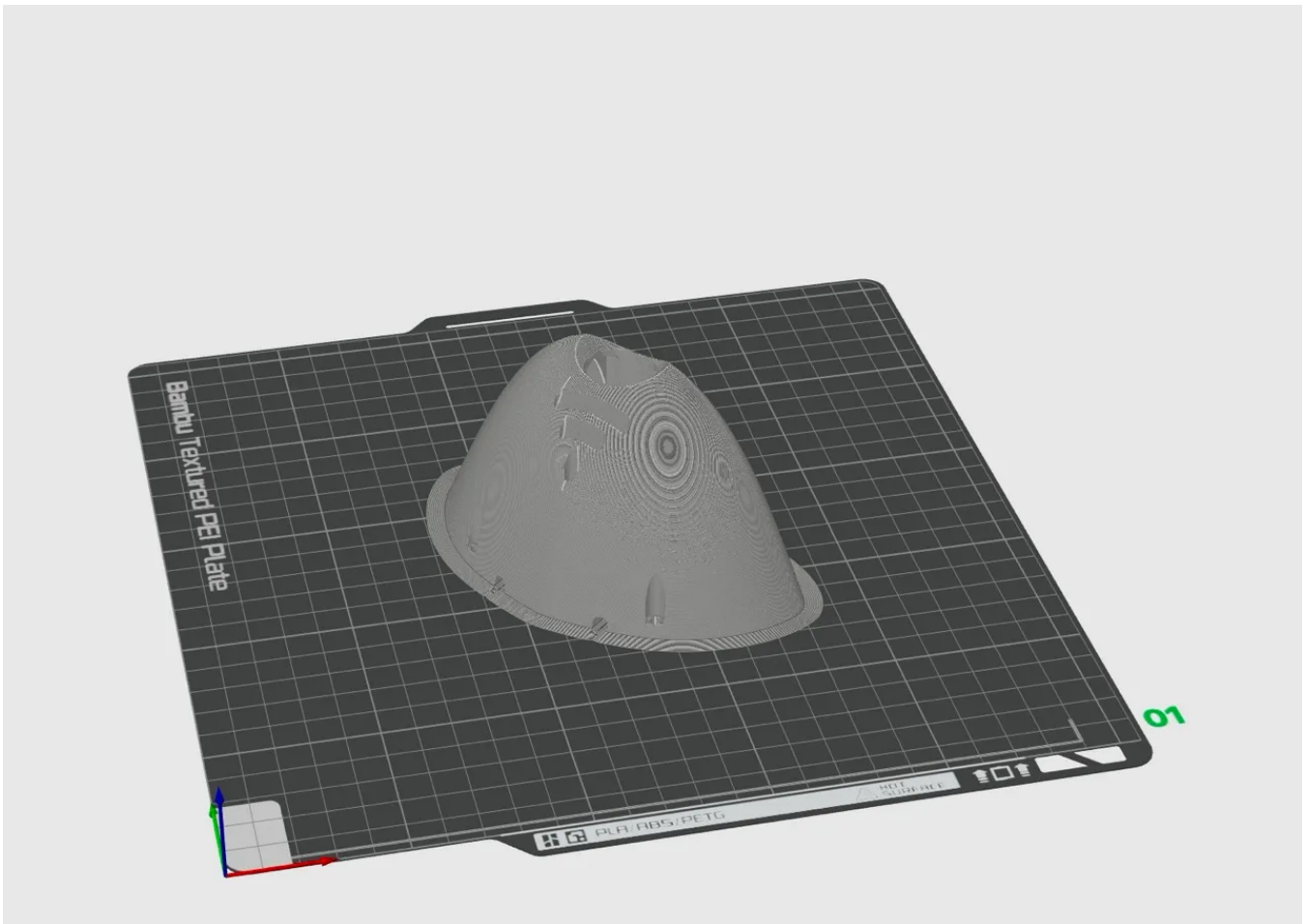


Tail Pusher | LW



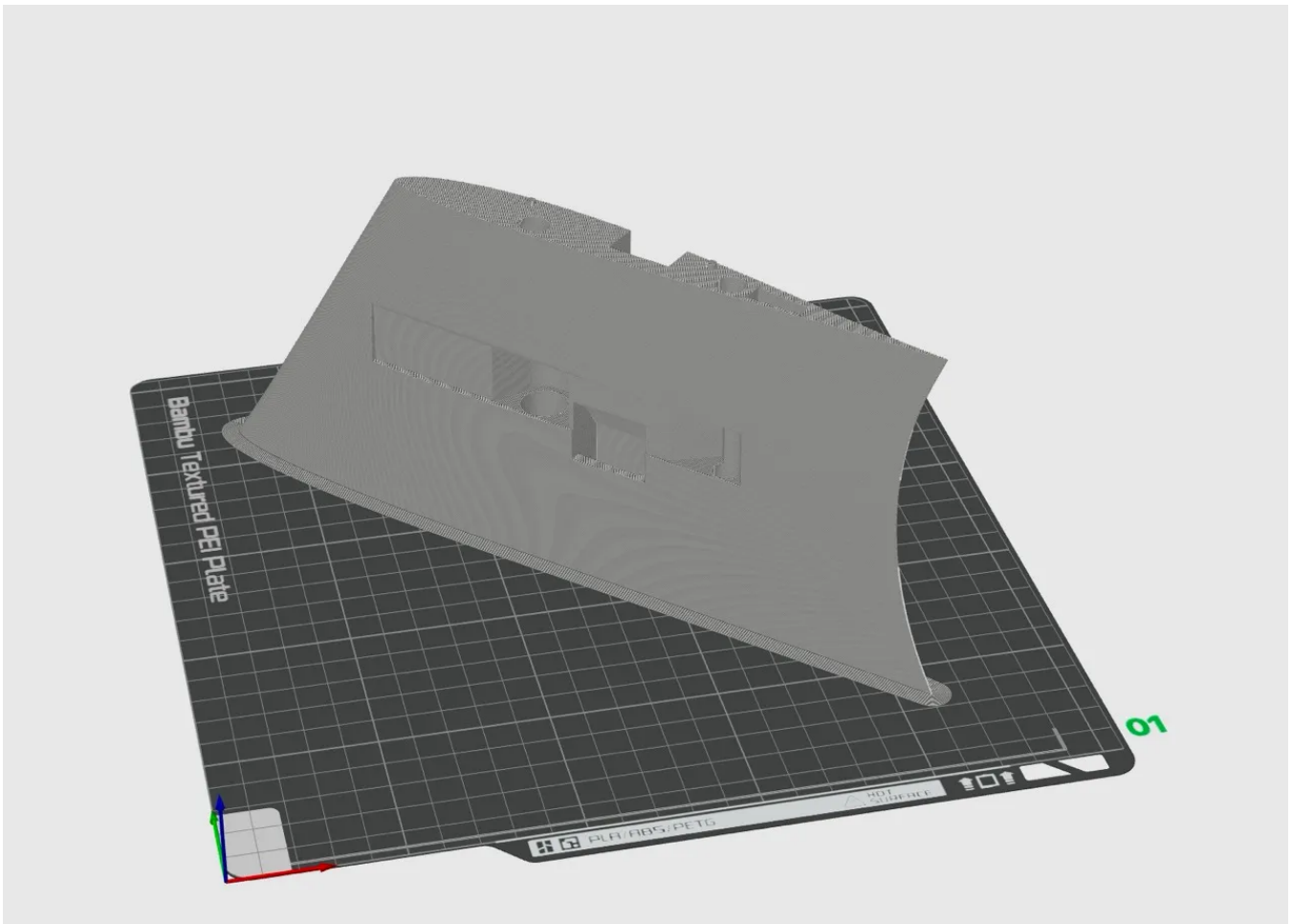
Nose | LW

- Multiple variants (15mm lens, 21mm lens) + editable STEP file available



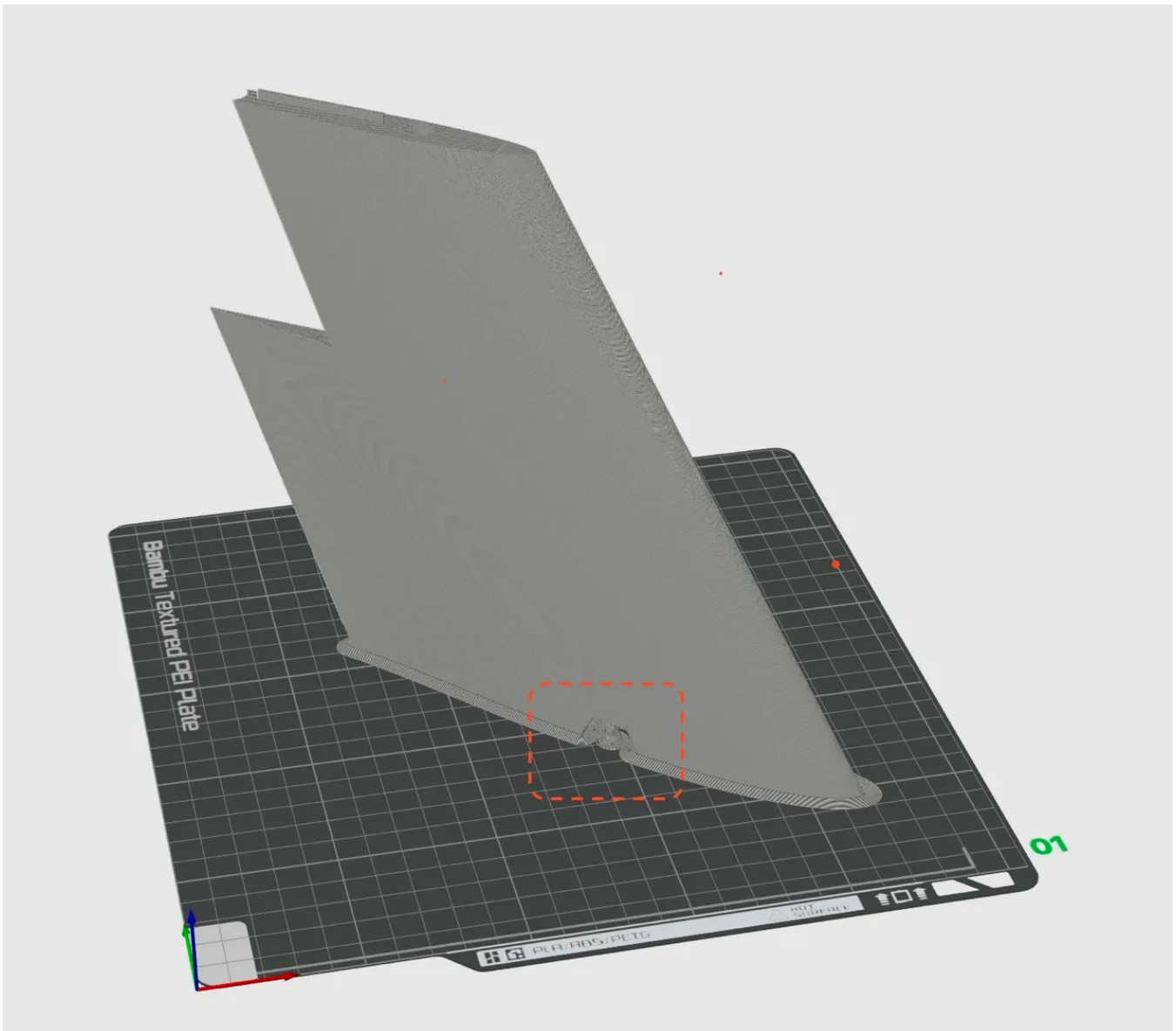
VTOL Section | LW | 2x Mirrored

- Mirror on X/Y and re-print
- Recommended: reinforce with 6% cubic, instead of 4%. This section does most of the work.



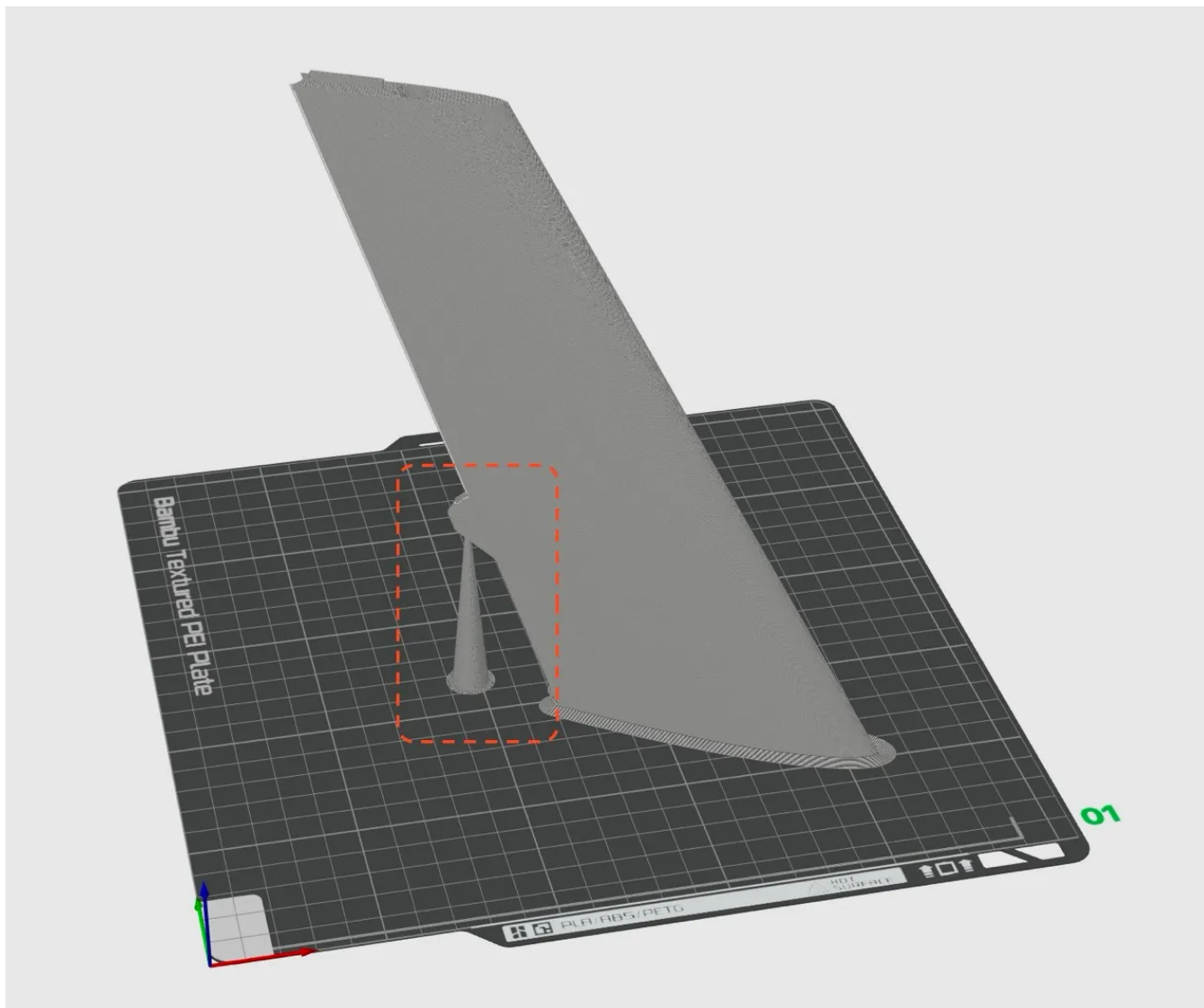
Wing Inner | LW | 2x Mirrored

- Required: Tree Support at 5% threshold angle (see red marking)



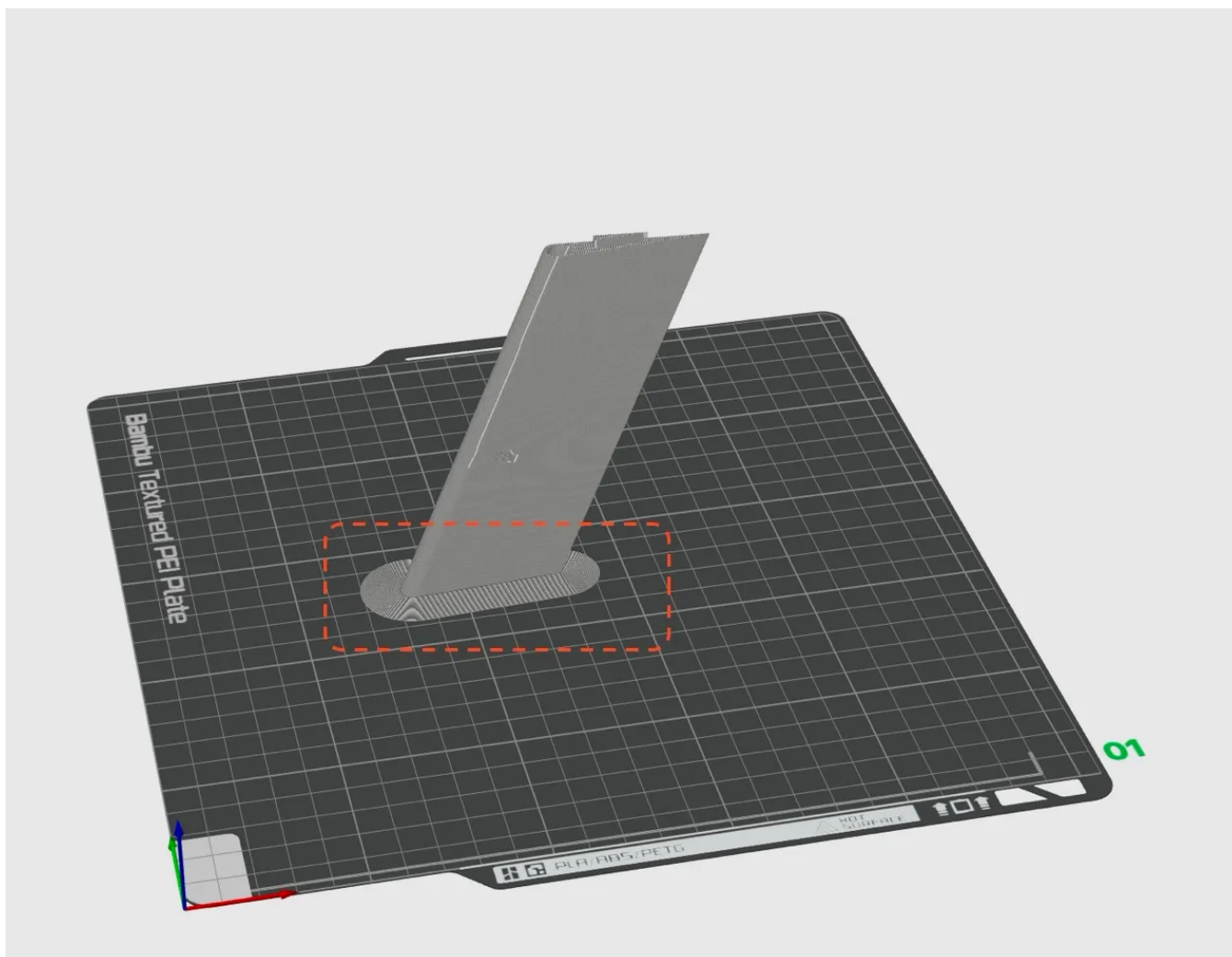
Wing Outer | LW | 2x Mirrored

- Required: Enable tree support at 15% threshold angle
- Recommended: Slow down by height
- Recommended: Increase brim count to 12 (helps with vibrations)



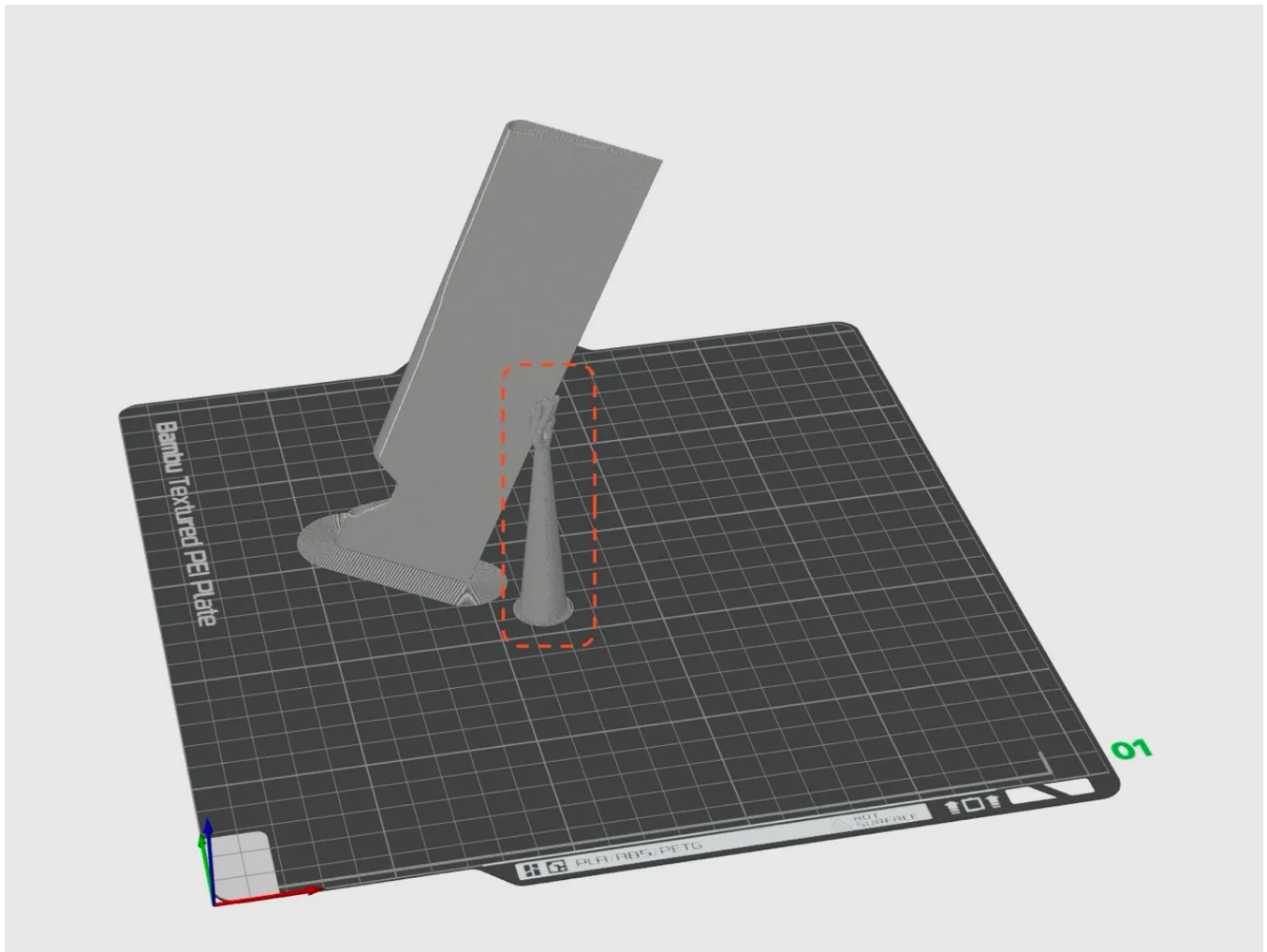
Elevon Inner | LW | 2x Mirrored

- Recommended: 12 brim



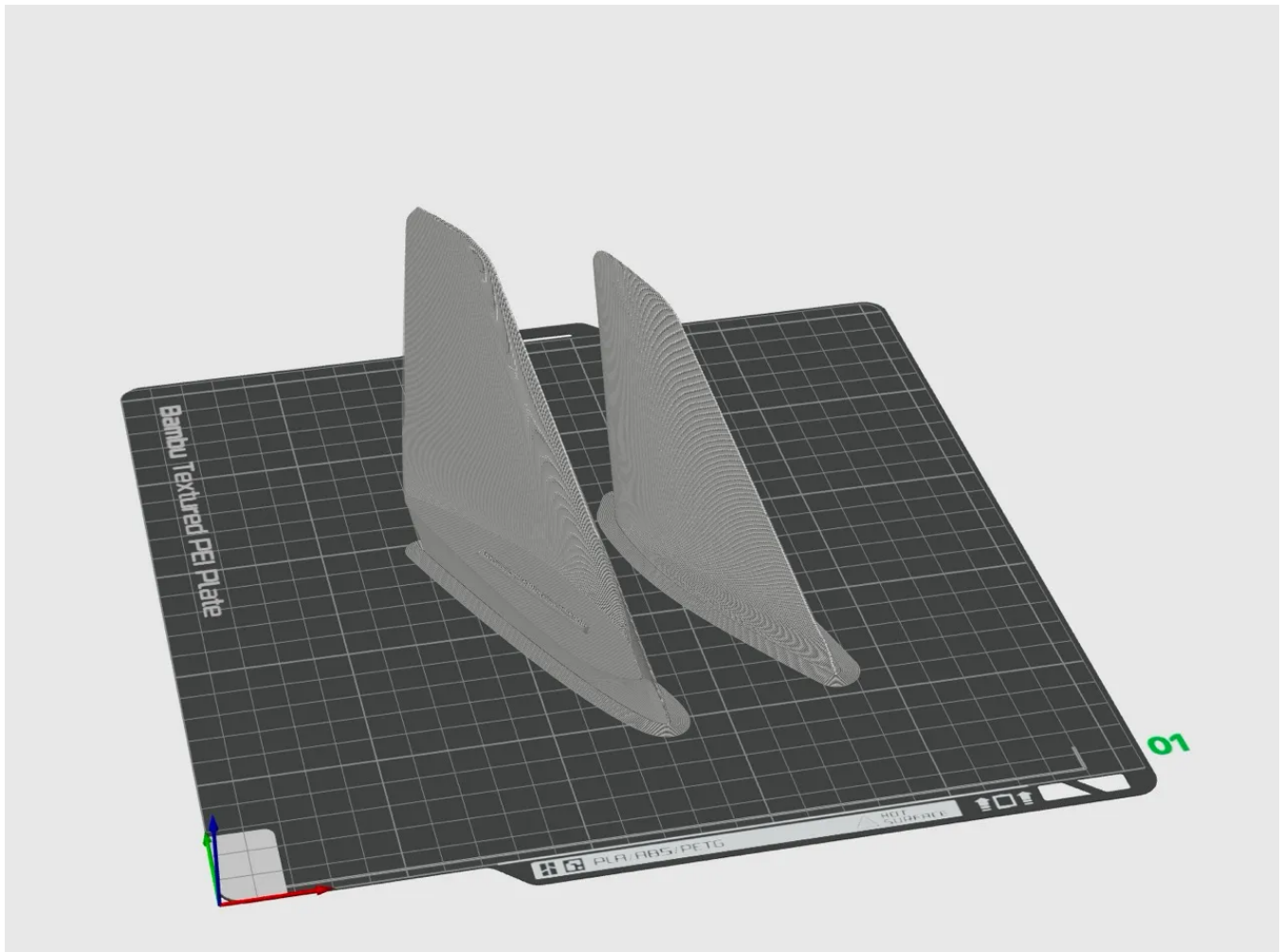
Elevon Outer | LW | 2x Mirrored

- Recommended: Draw a tree support using a sphere half-way on the trailing edge
- Recommended: 12 brim at 0.07 distance



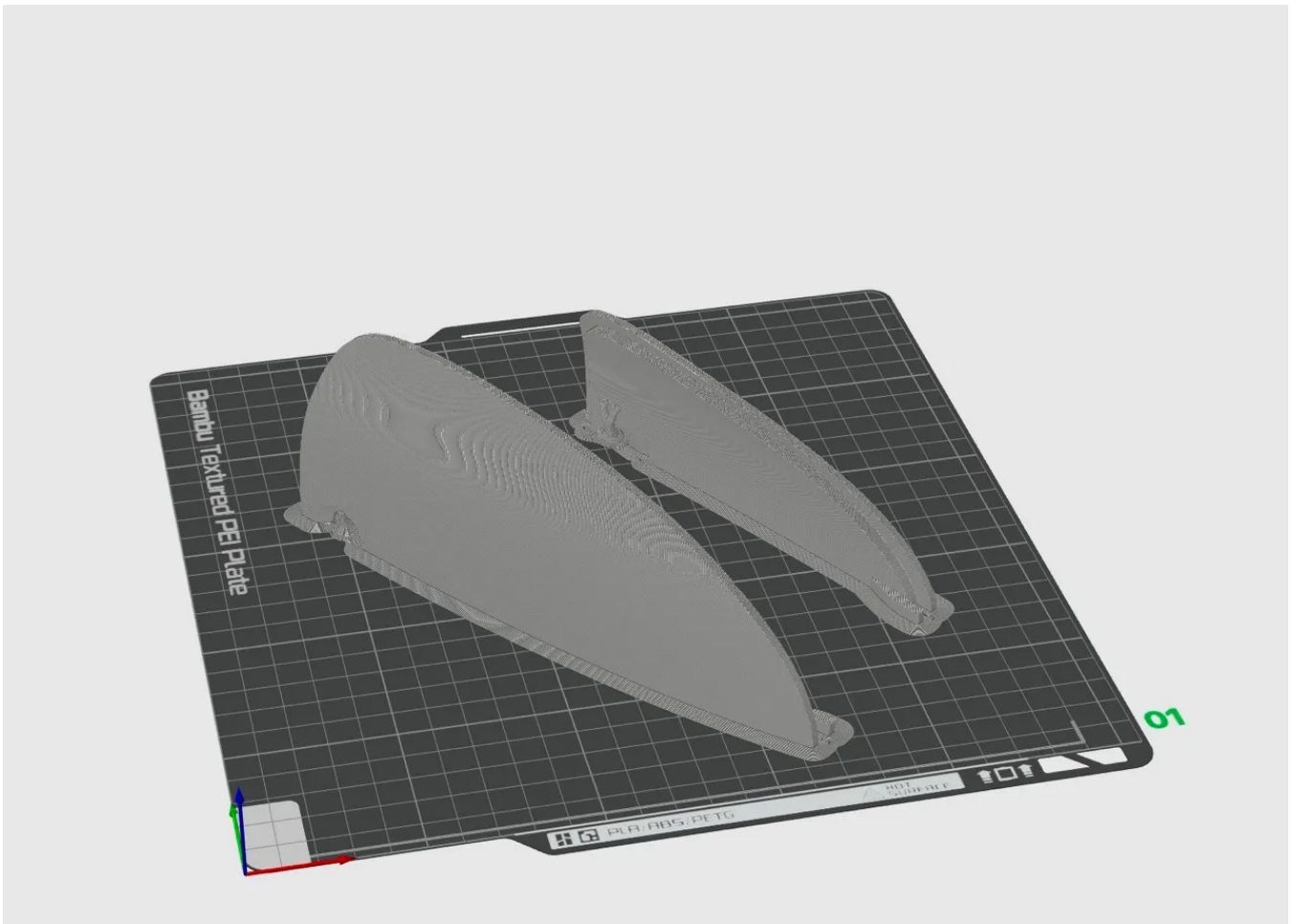
Winglet & Winglet Stabilizer | LW | 2x Mirrored

- Winglet Stabilizer comes in “Left” and “Right” due to your airframe’s serial number imprinted on the part. Print both, instead of mirroring one.
- 4 parts total



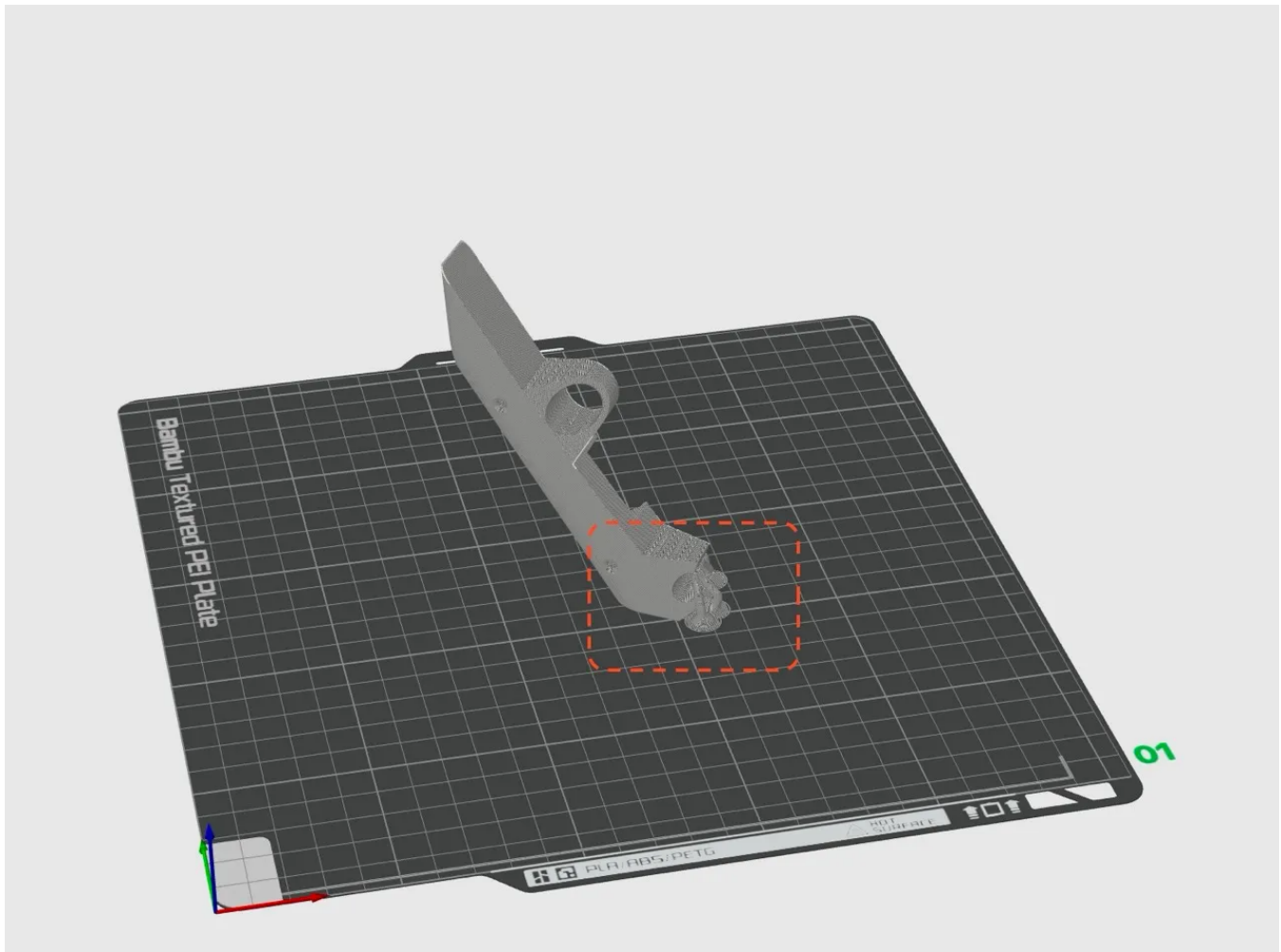
Hatch Front & Hatch Rear | LW | 2x Mirrored

- 4 parts total
- Recommended: Tree support ~15 deg threshold



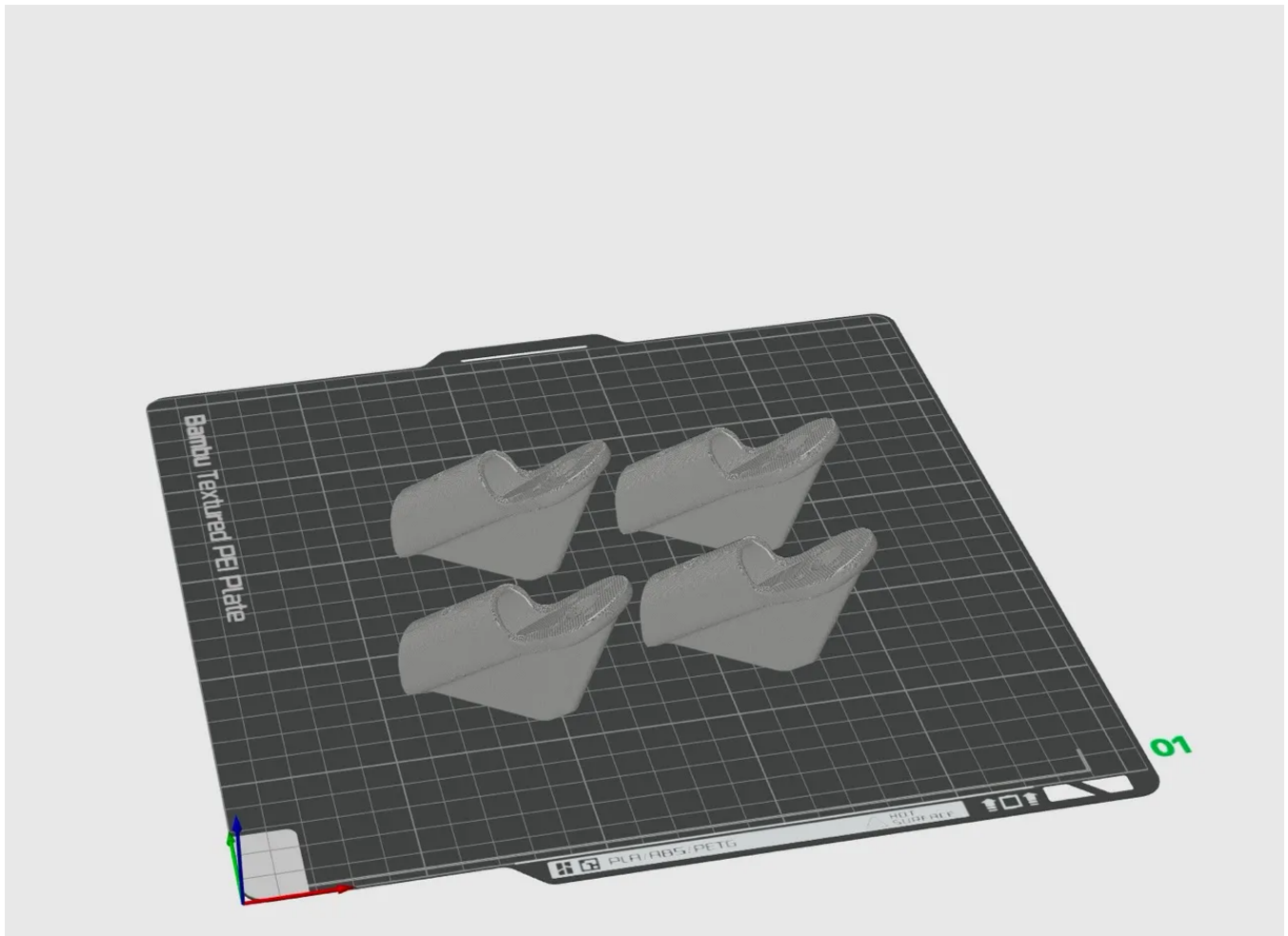
VTOL Spar Mount | PETG | 2x Mirrored

- Required: 3 Wall; 10% Gyroid;
- Required: “Lay on face” manually exactly as shown.
- Required: Tree support ~5 deg threshold
- Printed at angle for optimal layer adhesion



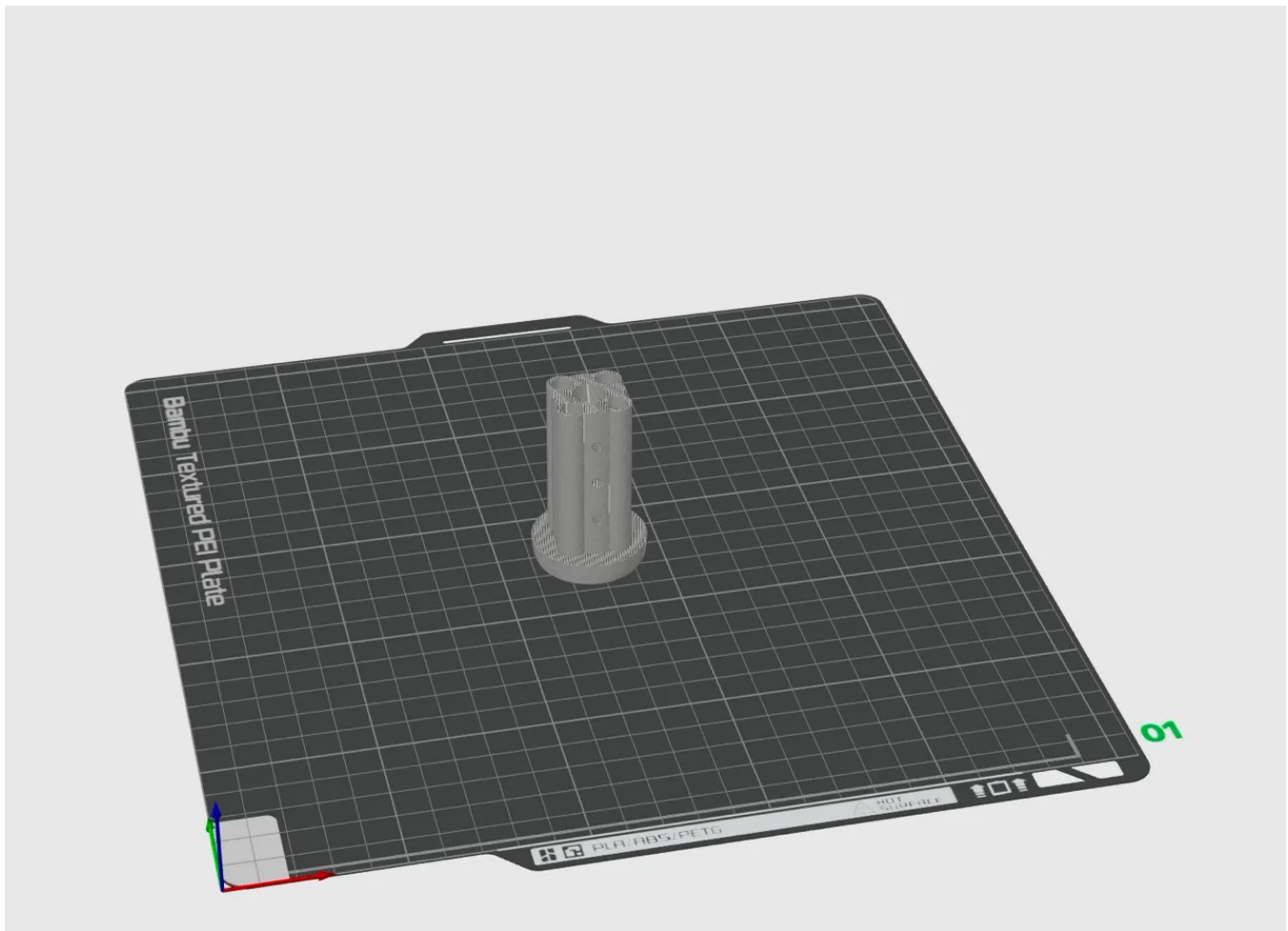
VTOL Motor Mount | PETG | 4x Identical

- Required: 3 Wall; 10% Gyroid;
- Required: “Lay on face” exactly as shown.
- Printed at angle for optimal layer adhesion



Pusher Motor Mount | PETG

- Required: ~2 Wall; 10% Gyroid;
- Required: Bottom Shell Layers: ~8 (better thermal resistance for the motor)



Remaining Parts | PETG (not all shown in image)

Tools:

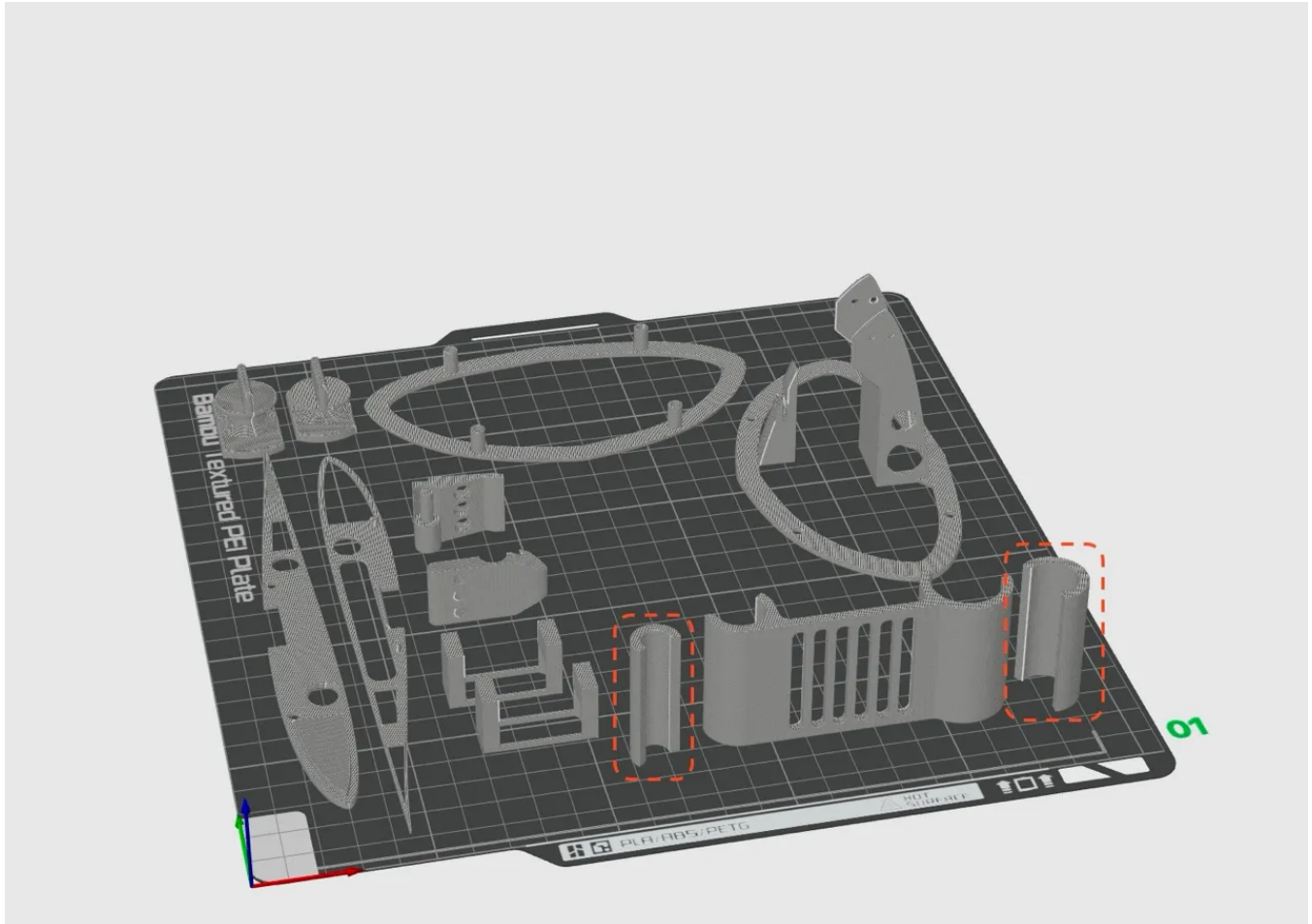
- VTOL Cutting Tool Main
- VTOL Cutting Tool Tip

Fuselage Section:

- Fuselage Connector Ring | 2x Mirrored (aligns fuselage front to fuselage mid)
- Fuselage Nose Plate
- Nose Plate
- Battery Tray
- Hatch Lock Front
- Hatch Lock Rear
- FC Mount (optional)
- FC Mount Dampener Front (TPU rubbery filament - see red marking)
- FC Mount Dampener Rear (TPU rubbery filament - see red marking)

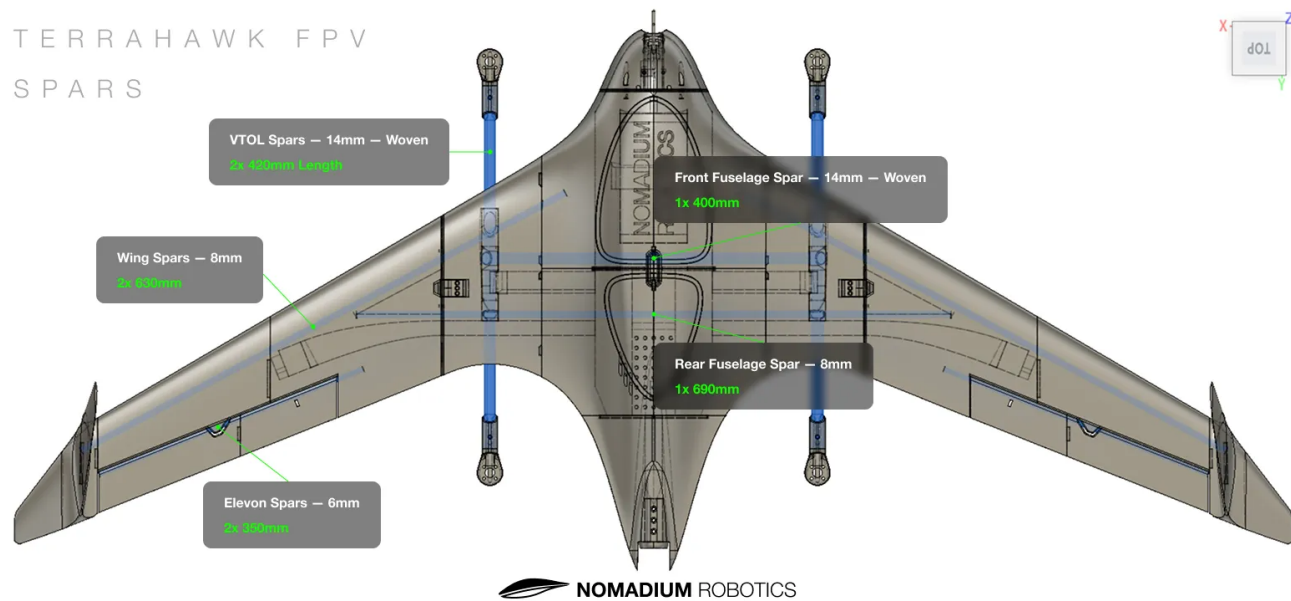
Wing Quick Release:

- VTOL Section Plate | 2x Mirrored
 - Wing Inner Plate | 2x Mirrored
 - Wing Clip Base | 2x
 - Wing Clip | 2x
- Other:
- Servo Mount | 2x



#3 ASSEMBLE THE AIRFRAME

TERRAHAWK FPV SPARS



TerraHawk FPV is built for stiffness, with wings you can still take off.

The two VTOL spars are held by the two fuselage spars in an H-frame. That is what keeps the quad rigid in hover.

The wings slide into the fuselage at an angle. That angle is what lets a wing this long stay stiff in fast forward flight.

Spar Cut List

Carbon fibre dust is a mechanical and respiratory irritant that may cause skin itching, eye irritation, and potential lung inflammation when inhaled during cutting, sanding, or machining. Use **safety goggles** and a **respiratory mask**, and preferably do it outdoors or in a professionally ventilated area.

Cut these from the stock you ordered in step 1. Lengths are rounded from CAD. Mark every cut before you make the first one.

Always cut spars a **bit shorter**, for anything other than the VTOL spars, to ensure the best fit between 3d printed segments. A 400mm spar should be ~395mm in reality.

Qty	Spar	Length	Stock
1	Rear Fuselage Spar	690mm	8mm pultruded
1	Front Fuselage Spar	400mm	14mm woven
2	Wing Spar	630mm	8mm pultruded
2	Elevon Spar	350mm	6mm pultruded
2	VTOL Spar	420mm	14mm woven

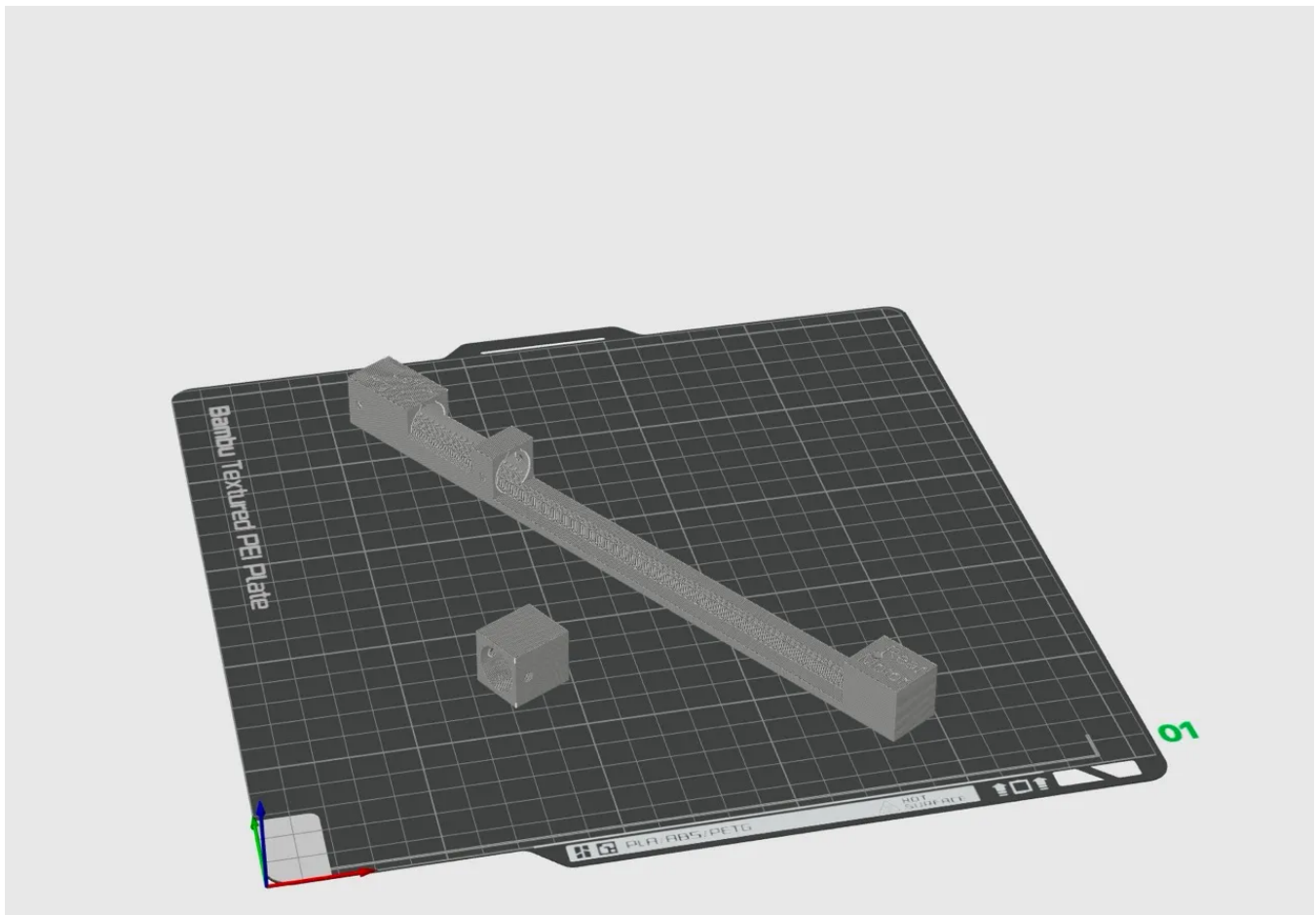
VTOL Spar Drilling

Use the printed VTOL Cutting Tools for every hole and the slot. They are in the STL kit and they set the positions for you. Put the tool on a flat surface, hold the spar steady, drill straight. Put cardboard underneath, because the bit comes out the other side.

Each VTOL spar gets:

- Two **vertical** holes, one at each tip, for the VTOL Motor Mounts.
- Two **horizontal** holes near the middle, for the VTOL Spar Mount.
- One **wire slot**, cut at 45 deg, so the motor wires can leave the spar and enter the fuselage.

The two spars are mirror images of each other. Cut a left and a right, not two of the same. The tool has “left”, “right” and “rear” markings imprinted on it.



Before You Glue Anything

CA glue is permanent. Run through this list before any of it goes together.

- [] Every part printed in the **correct material**, per the Segments Overview and the Part Orientation section
- [] Dry-fit the **whole airframe** with no glue: every segment, both wings, the nose cone
- [] Every **carbon spar** slides fully into its hole by hand, with no forcing
- [] Every **bolt hole** in the PETG parts takes its fastener

[] No part shows **layer separation** or a soft, crushable patch

[] Wings and elevons are **straight**. Spars should go all the way through them.

Do not glue the quick-release joints. The wings and the nose cone separate at the durable plates on purpose. Gluing them turns a field-serviceable aircraft into a large object that does not fit in your car.

Cyanoacrylate (CA) glue vapours may cause acute irritation to the eyes, nose, and mucous membranes. Use protective gear such as **safety goggles** and a **respiratory mask**.

Assembly Sequence

Six stages, in this order. Fuselage body, VTOL sections, finish the fuselage, wings, elevons, winglets. Each stage lists its parts first, so lay them out before you open the glue.

Two rules for every stage. Dry-fit every joint before you glue it. Keep glue off every carbon spar and away from every spar hole.

3.1 Fuselage Body

Parts: Fuselage Front, Fuselage Mid, 2× Fuselage Connector Ring

Stand **Fuselage Front** on its nose, on a flat surface.

Insert the two **Fuselage Connector Rings**, then test-fit **Fuselage Mid** onto them.

If the fit is good, glue it. If it is tight, clear away the excess filament first, then glue.

3.2 VTOL Sections

Parts: 2× VTOL Section, 2× VTOL Spar Mount, Front Fuselage Spar (14 mm), Rear Fuselage Spar, both VTOL Spars

Test-fit each **VTOL Spar Mount** by sliding a **VTOL Spar**, the **Front Fuselage Spar** and the **Rear Fuselage Spar** through it. Take them out again.

Insert the **Front Fuselage Spar** into **Fuselage Front** and centre it. Test-fit both **VTOL Sections** onto it at the same time. If the spar is too long, cut it shorter now. Take the sections off again.

Glue a **VTOL Spar Mount** into each **VTOL Section**.

Put the **Front Fuselage Spar** back in and centre it. Glue one **VTOL Section** to the fuselage body, then the other. Keep glue away from the spar where it passes through the joint.

3.3 Finish the Fuselage

Parts: Rear Fuselage Spar, Battery Tray, Fuselage Nose Plate, 2× VTOL Section Plate, Tail Pusher, 2× Wing Clip, 2× Wing Clip Base, Hatch Front (two segments and Hatch Lock Front), Hatch Rear (two segments and Hatch Lock Rear)

Insert the **Rear Fuselage Spar** into **Fuselage Mid** and centre it.

Glue the **Battery Tray** to **Fuselage Front**.

Glue the **Fuselage Nose Plate** into **Fuselage Front**.

Glue a **VTOL Section Plate** to each **VTOL Section**.

Glue the **Tail Pusher** to **Fuselage Mid**.

Push a **Wing Clip** into each **Wing Clip Base**. Glue each base into the cutout on its **VTOL Section**, with no glue on the clip. The clip must move freely.

Test-fit the **Hatch Lock Front** between the two **Hatch Front** segments. Glue the segments together with the lock in place, keeping glue away from the lock. It must rotate freely.

Repeat step 7 for the **Hatch Rear**.

The plates are the quick-release. Each plate you glue on here mates with a plate on a removable part: nose plate to nose cone, section plate to wing. Never glue plate to plate.

3.4 Wings

Parts, per wing: Wing Inner, Wing Outer, Wing Inner Plate, Wing Spar

Glue the **Wing Inner Plate** to **Wing Inner**.

Test-fit the **Wing Spar** all the way through **Wing Inner** and **Wing Outer**. Some friction is good. A spar you have to force will crack the part during assembly. If it binds, push it through a few times with a twisting motion, or run a drill lightly through the hole.

Glue **Wing Outer** to **Wing Inner**. Use the spar to check alignment while the glue is wet. If it starts to stick, pull it out with a twisting motion, wipe it clean, and wait for the joint to dry.

Push the **Wing Spar** in, all the way to the tip of **Wing Outer**.

Never glue the wing spar. Not to the wing, not to the plate, not anywhere. Keep glue away from the spar holes as well.

3.5 Elevons

Parts, per wing: Elevon Inner, Elevon Outer, Elevon Spar

The elevon spar is also the hinge. It runs through the elevon, through the hinge on Wing Outer, and into Wing Inner.

Test-fit the **Elevon Spar** through both elevon segments. Take it out again.

Glue **Elevon Inner** to **Elevon Outer**. Let it dry.

From the outer end of **Elevon Outer**, the longer segment, push the **Elevon Spar** in as far as the hinge cutout.

Line the elevon cutout up with the hinge on **Wing Outer**, then push the spar gently through the hinge.

Push the spar on until it seats in its hole in **Wing Inner**. The elevon should now swing freely.

3.6 Winglets

Parts, per wing: Winglet, Winglet Stabilizer

Slide the **Winglet** onto the outer tip of the **Elevon Spar**, then glue the **Winglet** to **Wing Outer**.

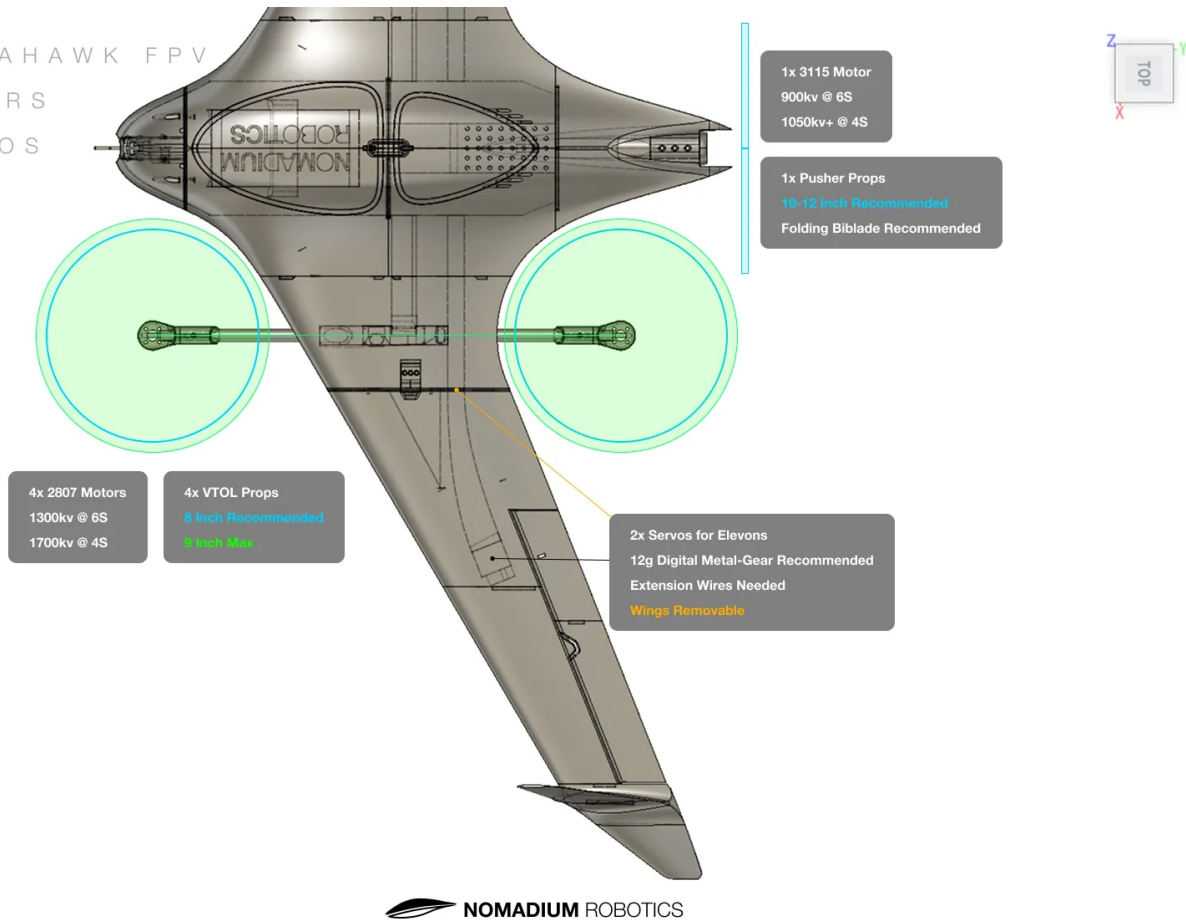
Push the **Wing Spar** the rest of the way, into the **Winglet**.

Glue the **Winglet Stabilizer** to the **Winglet**.

One wing done. Repeat 3.4 to 3.6 for the other wing, mirrored. Then stand back and have a look at what you have made.

#4 FIT THE ELECTRONICS

TERRAHAWK FPV
MOTORS
SERVOS



Everything goes in after the airframe is assembled. The removable wings and nose cone are there so you never have to wire through a glued joint.

4.1 Pusher Motor

Parts: Pusher motor (3115), Pusher Motor Mount, 4× M3 motor screws, 1 to 3 long M3 screws (30 mm or longer), thread-locker (optional)

Bolt the motor to the **Pusher Motor Mount** with four M3 screws. The mount has four sides, and two of them have screw holes running end to end. Turn the motor so its wires exit over one of those two sides.

Push the **Pusher Motor Mount** into the **Tail Pusher**, wires facing down. Feed the wires into the fuselage through the vent underneath.

Lock the mount in place with one to three long M3 screws, driven vertically.

Motor screws must not reach the windings. Hold each screw against the motor base and check its length before you drive it. A screw into the windings shorts the motor. A drop of thread-locker on each screw is good practice.

4.2 VTOL Motors

Parts: 4× VTOL motor (~2807), both drilled VTOL Spars, 4× VTOL Motor Mount, M3 screws (4 per motor, 2 long per spar mount, 2 long per motor mount), motor wire extensions, tape

Do one side at a time. Steps 2 to 11 are for one spar. Then do the other.

Extend the wires of all four motors so they reach from the spar tip to wherever the ESCs will sit in the fuselage. Motor wire is usually 18 AWG. Do not fit MR30 or any other connector yet. The spar is 14 mm across and a connector may not pass through it.

Slide the **VTOL Spar** into the **VTOL Spar Mount** under the **VTOL Section**.

Line up the two horizontal holes in the spar with the holes in the mount. A flashlight shone from behind helps you see through. The wire slot in the spar must line up with the cutout in the mount, or the wires have no way out.

Push a **VTOL Motor Mount** onto each tip of the spar, with the vertical holes lined up.

Tape the three wires of one motor together at the tips so they move as one. Push them into the spar through the motor mount, until they reach the wire slot.

Fish the wires out through the slot, then push them along the channel that joins the **VTOL Section** to the fuselage. They should end up inside the fuselage with enough slack to reach the ESC.

Repeat steps 5 and 6 for the motor at the other tip of the same spar.

Now the screws. Drive the two long horizontal M3 screws through the **VTOL Spar Mount** and the spar. By hand, no force. Keep sliding the wires back and forth a centimetre as you go. If they stop moving, the screw is on them. Back it out. Puncturing a wire can be catastrophic. Take your time. There is enough space inside for both the wires and the screw.

Drive the two long vertical M3 screws through each **VTOL Motor Mount** and the spar tip. Same rule with the wires.

Check the alignment on the two mounts on this spar. The landing feet should be perfectly vertical. If the front and rear mount disagree, the holes were drilled off. Align the mounts by eye and glue them instead of relying on the screws. Optionally, having the feet point inward into the fuselage with ~3 degrees may be beneficial for YAW control on competitive builds.

Bolt a motor into each **VTOL Motor Mount** with four M3 screws. Same rule as the pusher: check the screw length, and keep it out of the windings.

Repeat steps 2 to 11 for the other side.

Screws, not glue, for the spar mounts. You can glue the spar into its mount, but screws let you take the VTOL spars out again for repairs.

4.3 Servos and Control Horns

Parts, per wing: 9 g or 12 g servo, Servo Mount, servo extension lead, control horn, control rod

Fit the servo into its **Servo Mount**. If it will not fit, skip the mount and hot-glue the servo straight into the bay at step 6.

Test-fit the servo in the servo bay on **Wing Inner**. Orientation matters. The output gear sits toward the trailing edge, and the shaft points outboard, toward the wing tip.

Centre the servo with a servo tester, at 1500 μ s.

Extend the servo lead and feed it through the wire channel in **Wing Inner**.

Fit the servo arm so it stands perpendicular to the wing chord, and tighten its screw. You will not reach that screw again.

Glue the **Servo Mount** into the bay, with the servo fitted and its orientation checked.

Fit a control horn to the **Elevon**. Any hobby-shop horn works, or print one.

Link the servo arm to the horn with a control rod.

Tighten everything with the servo at centre and the elevon in line with the wing. Trim is done later during autopilot configuration.

Centre the servo before you glue it. Once the mount is in, the arm screw is buried. A servo glued in off-centre is an elevon you can never quite trim.

4.4 Nose Cone and VTX

Parts: Nose Cone, Nose Plate, your camera and video transmitter (VTX), M2 camera screws, 4× M3 screws

Mount the camera to the **Nose Plate** with M2 screws. Fix the VTX unit to the plate with hot glue or double-sided tape.

Cooling vents on the nose cone ensure some airflow during forward flight, but using an active fan on the air unit is highly recommended, especially for O3 Air Unit and similar systems, where an overheating VTX kills RC control as well.

Push the **Nose Plate** into the **Nose Cone**, seating the antenna, or antennas, in the cutouts on top of the cone. You can glue the plate to the cone, but you do not have to.

Fit the finished nose assembly to the front plate of the fuselage with four M3 screws.

Two nose cones are in the kit, for 15 mm and 21 mm camera lenses, plus an editable STEP file if yours is neither. Print the one that matches your camera.

Output Map

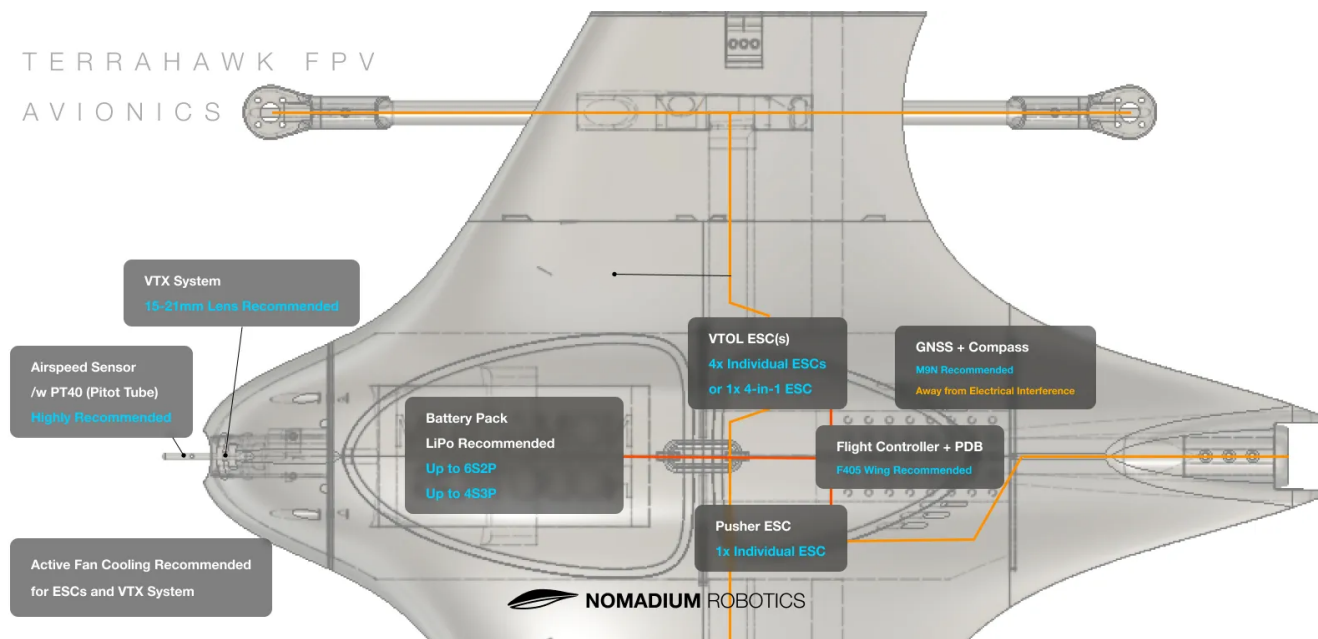
Wire it this way and the ArduPilot configuration in step 5 matches without changes.

Output	Goes to
S1	Lift motor, front right
S2	Lift motor, rear left
S3	Lift motor, front left
S4	Lift motor, rear right
S5	Pusher motor ESC
S6	Left elevon servo
S7	Right elevon servo

Check your board's output groups before you commit to this.

F405 and H7 boards group their outputs by hardware timer, and every output in a group must run the same protocol. All four lift motors need to sit in one DShot group, and the servos must **not** share a group with them. Move the assignments to suit your board, and keep the motor numbering intact.

Avionics



The layout above is our recommended starting point. The optional FC Mount and its TPU dampeners isolate the flight controller from motor vibration.

Component recommendations, with tested part numbers:

[Recommended Electronics List](#)

Centre of Gravity

Get this right before anything else. CG is critical for forward flight. ArduPilot will correct an error in CG, but it pays for it in flight time and stability, and it will not tell you it is doing so.

The Neutral Point (aerodynamic centre) is slightly forward of the **Rear Fuselage Spar**. CG should sit about **15% forward of the Neutral Point**, which puts it slightly aft of the **Front Fuselage Spar**.

Use the recommended electronics layout above as your starting point, then check the balance with the battery fitted, because the battery is the heaviest single item and the easiest one to slide.

Check it again with the flight battery in place. A CG measured on an empty airframe is not the CG you fly.

#5 CONFIGURE ARDUPILOT

New to ArduPilot? Start here instead. Every parameter set by hand, with the reasoning, in about 30 minutes.

[ArduPilot QuadPlane Setup: Beginners Guide](#)

Propellers OFF for everything in this step. You will be arming motors and running output tests on the bench. Put them back on outdoors, after the checks pass.

TerraHawk FPV is a **4+1 quadplane**: four lift motors for vertical take-off and landing, one pusher for forward flight, two elevons for control.

Firmware and Frame

Three things, in this order. Do them before anything else, because most of the table below does not exist until they are done.

Flash ArduPlane. Find your exact board on firmware.ardupilot.org and load the Plane firmware. "F405" is a chip, not a board. The electronics list gives the firmware target name for every board we recommend.

Turn on QuadPlane. In **CONFIG** → **Full Parameter Tree**, set Q_ENABLE to **1**. Write Params, then reboot. The rest of the Q_ parameters do not appear until you do.

Set the frame. Q_FRAME_CLASS to **1** (Quad) and Q_FRAME_TYPE to **1** (X). The four lift motors sit at four corners, so it is an X.

Refresh Params after the reboot. If you cannot find Q_FRAME_CLASS, this is why. It is not missing, the ground station just has not asked for it yet.

Key Values

Now work down this table. Set each one in **CONFIG** → **Full Parameter Tree**, then **Write Params**.

Parameter	Value	Why
Q_ENABLE	1	Turns on QuadPlane. Write and reboot before the other Q_ params appear
Q_FRAME_CLASS / Q_FRAME_TYPE	1 / 1	Quad, X layout
Q_M_PWM_TYPE	5 (DShot300)	Survives the long signal runs out to the VTOL spars. Pick it from the dropdown, not by number
SERVO1..4_FUNCTION	33, 34, 35, 36	Lift motors 1-4, per the output map in step 4
SERVO5_FUNCTION	70	Pusher throttle
SERVO6/7_FUNCTION	77 / 78	Elevon left / right. No separate aileron or elevator
AIRSPPEED_MIN / CRUISE / MAX	16 / 20 / 30 m/s	Starting points for a ~2kg build
Q_ASSIST_SPEED	14 m/s	Lift motors catch the wing below this speed. Do not disable this
Q_ASSIST_ANGLE	45 deg	Also catches an attitude upset before it becomes a spin
Q_TRANSITION_MS	5000 ms	Leave at 5 seconds for the maiden. Tighten toward 3000 once you trust it

Parameter	Value	Why
Q_M_BAT_VOLT_MAX / MIN	25.2 / 19.8 (6S)	Hover thrust stays steady as the pack drains. Needs a working BATT_MONITOR
Q_LAND_FINAL_ALT / SPD	3 m / 0.3 m/s	Gentle enough for printed landing feet
Q_OPTIONS	Level Transition, ThrLandControl, EnableLandReposition	Tick the checkboxes, never type the number

Check motor order and direction before you fly. This is the step people skip, and it is the step that breaks airframes. **SETUP** → **Optional Hardware** → **Motor Test**, props off, one motor at a time. Mission Planner tests A, B, C, D **clockwise from front right**, not in motor-number order. Full procedure in the beginners guide.

Still To Do Before You Fly

The airframe is configured. These are normal ArduPilot jobs, and the beginners guide lists them in order:

- **Radio** - RC calibration, and flight modes on a switch. At minimum: QSTABILIZE, QHOVER, FBWA, QLAND.
- **Sensors** - accelerometer calibration on a level surface, compass calibration away from metal, airspeed sensor setup. The airspeed sensor is calibrated before every flight, see **Airspeed Sensor Calibration** in step 6.
- **Battery monitor** - BATT_MONITOR, then calibrate voltage against a multimeter.
- **Failsafes** - radio failsafe, battery failsafe, GPS failsafe. Set these before the maiden, not after the first scare.
- **Tuning** - the defaults will hover. They will not hover well. Read the QuadPlane VTOL tuning guide once you have a few flights in the log.

#6 FIRST FLIGHT

Maiden Flight Checklist

Do not skip to this list.

If the CG is wrong or a motor spins backwards, nothing below saves you. Work through it in order, on the day, out loud.

On the bench

- [] CG checked and correct (see **Centre of Gravity** in step 4). Get this right first. Everything else is downstream of it.
- [] All four lift motors spin in the right place, in the right direction
- [] Propellers match motor direction, and all are tight
- [] Elevons move the correct way in FBWA
- [] Radio calibrated, flight modes on a switch, failsafes set
- [] Accelerometer and compass calibrated
- [] Q_M_THST_HOVER set to **0.25** for the first flight. It learns the real value as you hover.
- [] Battery charged, secured, and CG rechecked with it in place

At the field

- [] Low wind. Open space. A spotter. No people downwind.
- [] Props on now
- [] Pitot loosely covered before you power on, then uncovered. See **Airspeed Sensor Calibration** below
- [] GPS fix, HDOP under 1.5, EKF happy
- [] Arm in **QSTABILIZE**. Lift to one metre. Hover thirty seconds. Land. Disarm.
- [] Happy with the hover? Repeat in **QHOVER** to test the altitude controller.
- [] Climb to a safe height in QHOVER. Switch to **FBWA**. Feed in throttle. Let it accelerate and transition.
- [] Fly a gentle circuit. Do not tune, do not push it. Get a feel for it.
- [] Switch back to **QHOVER**, throttle to centre, and land.
- [] Download the log. Read it before flight two.

Let Augury turn your logs into insight.

Drop your .bin into Augury and you get the flight analysed properly. It will catch the tuning issues you cannot feel through the sticks. Available at:

- <https://augury.nomadiumrobotics.com>
- <https://hangar.nomadiumrobotics.com/my-hangar>

Airspeed Sensor Calibration

ArduPilot uses airspeed to decide when the wing is flying and when the lift motors should step in. A wrong reading makes that decision wrong, at the worst possible moment.

There are two separate jobs here. The **zero offset** is done before every flight. The **ratio** is learned once, on a calm day, and then left alone.

Before every flight: the zero

The sensor zeroes itself while the board boots, and stores the result in ARSPD_OFFSET. Any wind blowing into the pitot at that moment gives you a false zero, and a false zero follows you into the air.

Cover the pitot loosely before you power on. A bag, a bit of foam, a sock. Enough to keep the wind out.

Do not seal it, and do not squeeze the tube. A sealed tube traps a pressure that the sensor then calls zero. Loose is the whole point.

Power on, let the aircraft finish booting, then uncover the pitot.

Missed it, or the wind picked up? Redo it from your ground station's **Preflight Calibration** action, with the pitot loosely covered again.

Let it warm up first. If the aircraft has come out of a cold car, give the electronics a minute before a manual recalibration. A sensor that is still finding ambient temperature drifts while you calibrate it.

DLVR sensors skip this. The DLVR-class sensors on the advanced route do not need the boot-time zero. MS4525-class sensors do. If you are not sure which you fitted, cover the pitot anyway. It costs you nothing.

Once: learn the airspeed ratio

ARSPD_RATIO turns the pressure the sensor reads into a speed. The default of **2.0** is a starting point, not your aircraft. ArduPilot can learn the right value by comparing indicated airspeed against GPS ground speed, and that comparison only works when it can see you turning.

Fly this as its own flight, once the maiden is behind you.

[] Pick a calm day. You are asking the autopilot to separate airspeed from wind, so give it as little wind as possible

[] Set ARSPD_AUTOCL to **1**

[] Take off, transition, and climb to a safe height. Around 100 m is right

[] Circle in **FBWA** for about five minutes. Keep turning, five or six full circles

[] Land

[] Set ARSPD_AUTOCL back to **0**, and **Write Params**

[] Check that ARSPD_RATIO has moved off 2.0. That number is yours now, and it stays

Fixed-wing LOITER does the circling for you. Set a loiter point, switch to **LOITER**, and the aircraft turns steadily on its own while you watch the telemetry. That is **LOITER**, the fixed-wing mode, not **QLOITER**.

Turn auto-cal off again. ArduPilot saves a new ratio every two minutes, but only when its estimate has moved by more than 5%. Leave ARSPD_AUTOCL on through long straight flights and it keeps adjusting on data that has no turns in it. The reported airspeed then walks away from the truth, and everything downstream of it, Q_ASSIST_SPEED included, goes with it.

Changed the pitot tube? Put ARSPD_RATIO back to **2.0** and fly the calibration again. The ratio belongs to the tube, not to the board.

Going deeper. Every ArduPilot reference link, plus the full fresh-configuration walkthrough:

[ArduPilot QuadPlane Setup: Beginners Guide](#)